Problem J. The Phantom Menace

Input file:	`standard input`
Output file:	`standard output`
Time limit:	4 seconds
Memory limit:	1024 megabytes

Putata has brought his newest string problem to this contest. You are given two string sequences A, B, each of the sequences contains exactly n strings, and all strings have a length of m. You are asked to reorder the strings so that concatenation of the strings in the two sequences are cyclic isomorphic after concatenation.

Formally, you should choose two permutations p, q of $1, 2, \ldots, n$, so that $A_{p_1} + A_{p_2} + \cdots + A_{p_n}$ and $B_{q_1} + B_{q_2} + \cdots + B_{q_n}$ are cyclic isomorphic. String $R = S + T$ satisfy that for $i \le |S|, R_i = S_i$, otherwise $R_i = T_{i-|S|}$. Two strings S, T are said to be cyclic isomorphic if and only if there exists an integer d, such that $S_i = T_{((i+d) \mod |S|)+1}$ for all $1 \le i \le |S|$, and $|S| = |T|$.

Please help Budada to find p and q, or report that there is no such p, q.

Input

The first line contains one integer t ($1 \le t \le 10^6$), denoting the number of test cases.

For each test case, the first line contains two integers n, m ($1 \le n, m \le 10^6, 1 \le n \cdot m \le 10^6$).

Then n line follows, the i-th of which contains one string A_i ($|A_i| = m$).

Then n line follows, the i-th of which contains one string B_i ($|B_i| = m$).

It is guaranteed that all input strings only contain lowercase English letters.

It is also guaranteed that the sum of $n \cdot m$ over all test cases does not exceed 10^6.

Output

For each test case, if permutation p and q exists, output them in two lines, and the elements in one permutation are seperated by spaces. Otherwise output -1 in one line.

Example

standard input	standard output
2 3 3 abc ghi def bcd efg hia 1 3 abc def	1 3 2 1 2 3 -1